

CHANGWE MUSONDA

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EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science in Computer Engineering

May 2026

GPA: 3.56

Crafton Hills College

Associate of Science in Mathematics

May 2023

GPA: 3.67

WORK EXPERIENCE

Team Pegasus (Lockheed Martin Sponsored)

Pomona, CA

Autonomy Engineer → Autonomy Lead (Senior Capstone)

Jan 2025 – May 2026

- Led a team of 7 engineers to develop and integrate an end-to-end UAV autonomy system using **ROS 2**, **PX4**, and **Pixhawk**, supporting both simulation and real-world deployment workflows
- Developed modular **C++** and **Python** software components for sensor fusion, state estimation, and closed-loop control on **ARM**-based embedded systems (Jetson)
- Integrated **LiDAR**, **stereo cameras**, and **IMUs** into a unified perception pipeline supporting **SLAM**-based localization and real-time field performance
- Diagnosed hardware and software integration failures using telemetry logs and root-cause analysis, improving system reliability across lab and flight test environments
- Collaborated with cross-functional teams to support system integration, test readiness, and iterative validation from lab to field deployment

Autonomous Vehicle Laboratory | autovehiclelab.com

Pomona, CA

Research Assistant

May 2025 – Present

- Developed real-time robotic software in **C++** and **Python** for localization, mapping, and control on **ARM**-based Linux systems within **SLAM**-based autonomous platforms
- Integrated **LiDAR**, **stereo cameras**, and **Xsens IMUs** within multi-sensor fusion pipelines, improving perception accuracy and stability across real-world operating conditions
- Diagnosed and resolved failures across distributed **ROS 2** systems during lab-based integration, debugging, and testing workflows
- Prepared system documentation including architecture design, calibration procedures, and validation workflows to support reproducibility and team-wide integration

PROJECTS

Project DOLPHIN (AUV Swarm Autonomy)

Jan 2026 – Present

- Designed the **behavior tree** autonomy architecture for a swarm of submersible AUVs, running at 10 Hz on a **Raspberry Pi 5** and commanding a **Pixhawk 6X** flight controller via **MAVSDK** in **C++17**
- Architected a priority-ordered **fault handling** system covering flight controller link loss, battery management, leak detection, perception degradation, and acoustic communications loss
- Defined the autonomy/control/perception interface: the Pi proposes intent, the Pixhawk owns execution, and perception from a **Jetson Orin Nano** is treated as advisory input with **confidence gating** to prevent bad detections from driving control
- Designed a comms loss policy treating **acoustic link loss** as expected operating behavior, enabling fully autonomous operation without dependence on inter-vehicle communication

Deep Differentiable Neural Planner for Egocentric Navigation

Jan 2026 – May 2026

- Trained and benchmarked four differentiable planning architectures (**CALVINConv2d**, **CALVINConv3d**, **VIN**, **GPPN**) on custom egocentric and non-egocentric maze datasets, extending a CVPR 2022 paper to evaluate model behavior across dataset variants and identify performance gaps
- Modernized a legacy **PyTorch** research codebase for current library compatibility, debugging GPU/CPU tensor handling and CUDA dependencies to establish a reproducible ML experiment pipeline

Unmanned Ground Vehicle SLAM System

Dec 2025 – Present

- Developed a **ROS 2**-based **SLAM** pipeline integrating mapping and localization to support autonomous navigation in unstructured environments
- Improved localization robustness by analyzing failure cases during testing and refining system parameters to increase stability under sensor noise and environmental variation

A* Path Planning on FPGA (Verilog + Python)

Aug 2025 – Dec 2025

- Implemented a hardware-accelerated path planning algorithm in **Verilog**, enabling parallel execution for faster computation on resource-constrained embedded processors
- Validated hardware output against **Python** reference models across structured test scenarios, measuring execution time improvements and resource utilization

Medical Image Segmentation (PyTorch)

Aug 2025 – Dec 2025

- Developed **PyTorch** pipeline for cardiac MRI segmentation with reproducible training workflows
- Improved segmentation consistency through preprocessing and metric-driven evaluation

TECHNICAL SKILLS

Languages: C/C++, Python, Verilog

Tools: ROS 2 (Humble), PX4, Pixhawk, MAVSDK, PyTorch, Git, Linux

Systems: ARM-Based Embedded Systems (Jetson, Raspberry Pi), Behavior Tree Architecture, Sensor Fusion, SLAM, Xsens IMU, System Integration & Testing, Embedded Linux